NO.	ANSWER	SOLUTION (PART A)
1.	D	$\lim\limits_{x\to -2} 2 = 2$
2.	C	$\dfrac{d}{dx}\left[\dfrac{(x+1)^2}{x^2+1}\right]$ $= \dfrac{(x^2+1)(2x+2)-(x+1)^2(2x)}{(x^2+1)^2}$
3.	B	$y = 7x^4 + 8x^{-2}$ $\dfrac{dy}{dx} = 28x^3 - 16x^{-3}$ $\dfrac{d^2y}{dx^2} = 84x^2 + 48x^{-4}$
4.	C	$f(x) = 3x^4 + 4x^3$ $f'(x) = 12x^3 + 12x^2$ $f'(x) = 0$ $12x^3 + 12x^2 = 0$ $12x^2(x+1) = 0$ $x = 0$ or $x = -1$ When $x = 0$, $f(0) = 3(0)^4 + 4(0)^3 = 0$ $x = -1, f(-1) = 3(-1)^4 + 4(-1)^3 = -1$ Critical points are $(-1,-1)$ and $(0,0)$
5.	B	$\lim\limits_{x\to 2} \dfrac{x-2}{4x^2-16}$ $= \lim\limits_{x\to 2} \dfrac{(x-2)}{4(x^2-4)}$ $= \lim\limits_{x\to 2} \dfrac{(x-2)}{4(x-2)(x+2)}$ $= \lim\limits_{x\to 2} \dfrac{1}{4(x+2)}$ $= \dfrac{1}{4(2+2)} = \dfrac{1}{16}$

NO.	ANSWER	SOLUTION (PART A)
6.	D	$\frac{d}{dx}\left(x^2-x\right)^3$ $=3\left(x^2-x\right)^2(2x-1)$
7.	C	$\frac{d}{dx}\left(\cos^2 3x\right)$ $=2\cos 3x(-\sin 3x)(3)$ $=-6\cos 3x\sin 3x$
8.	C	$\lim\limits_{x\to-1}\left[f(x)+g(x)\right]=\lim\limits_{x\to-1}f(x)+\lim\limits_{x\to-1}g(x)$
9.	B	$y=\frac{3e^{1-2x}}{4}$ $\frac{dy}{dx}=\frac{3}{4}\left[e^{1-2x}\right](-2)$ $=\frac{-6e^{1-2x}}{4}$
10.	A	$y=x^4-8\sqrt{x}$ $y'=4x^3-8\left(\frac{1}{2}x^{-\frac{1}{2}}\right)$ $=4x^3-4x^{-\frac{1}{2}}$ $y''=12x^2-4\left(-\frac{1}{2}x^{-\frac{3}{2}}\right)$ $=12x^2+2x^{-\frac{3}{2}}$ $=12x^2+\frac{2}{x\sqrt{x}}$
11.	A	$f(x)=4x^3-48x+24$ $f'(x)=12x^2-48$ $f''(x)=24x$ $x=2$, $f''(2)=24(2)$=48 >0 (minimum pt) $x=-2$, $f''(-2)=24(-2)$=-48 >0 (maximum pt)

<table>
<tr><th>NO.</th><th>ANSWER</th><th>SOLUTION (PART A)</th></tr>
<tr><td>12.</td><td>D</td><td>

$y = -x^3 + 3x^2$

$y' = -3x^2 + 6x$

$y' = 0, \quad -3x(x-2) = 0$

$x = 0$ or $x = 2$

Interval	$(-\infty, 0)$	$(0,2)$	$(2,\infty)$
$f'(x)$	$-ve$	$+ve$	$-ve$
Conclusion	decreasing	increasing	Decreasing

</td></tr>
<tr><td>13.</td><td>B</td><td>

$\lim\limits_{x\to 13} \frac{(x+7)(f(x)+1)}{g(x)}$

$= \frac{\lim\limits_{x\to 13}(x+7)\left(\lim\limits_{x\to 13} f(x) + \lim\limits_{x\to 13} 1\right)}{\lim\limits_{x\to 13} g(x)}$

$= \frac{(13+7)(-3+1)}{5}$

</td></tr>
<tr><td>14.</td><td>D</td><td>

$\lim\limits_{x\to 0}\left(\frac{12}{x} + \frac{3x-12}{x}\right)$

$= \lim\limits_{x\to 0}\left(\frac{12+3x-12}{x}\right)$

$= \lim\limits_{x\to 0}\left(\frac{3x}{x}\right)$

$= \lim\limits_{x\to 0}(3) = 3$

</td></tr>
<tr><td>15.</td><td>C</td><td>

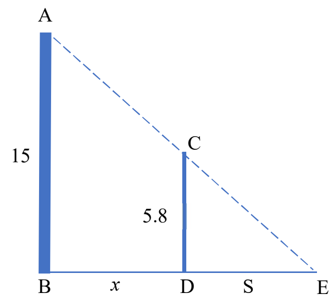
$\frac{d}{dx}\left(e^{-3x}\right) = \frac{-3}{e^{3x}}$

</td></tr>
</table>

NO.	ANSWER	SOLUTION (PART B)
1.	C	$\log_{5-x}\left(x^2-6x+65\right)=2$ $x^2-6x+65=\left(5-x\right)^2$ $x^2-6x+65=25-10x+x^2$ $4x=-40$ $x=-10$
2.	C	$\dfrac{13}{\sqrt{3}\left(5-2\sqrt{3}\right)}=\dfrac{13}{5\sqrt{3}-6}$ $=\dfrac{13}{5\sqrt{3}-6}\times\dfrac{5\sqrt{3}+6}{5\sqrt{3}+6}$ $=\dfrac{65\sqrt{3}+78}{75-36}$ $=\dfrac{65\sqrt{3}+78}{39}$ $=2+\dfrac{5}{3}\sqrt{3}$
3.	B	$\dfrac{(1-a)^{\frac{1}{2}}}{(1-a)^{-\frac{1}{2}}-(1-a)^{\frac{1}{2}}}=\dfrac{(1-a)^{\frac{1}{2}}}{\dfrac{1}{(1-a)^{\frac{1}{2}}}-(1-a)^{\frac{1}{2}}}$ $=\dfrac{(1-a)^{\frac{1}{2}}}{\dfrac{1-(1-a)}{(1-a)^{\frac{1}{2}}}}$ $=\dfrac{(1-a)^{\frac{1}{2}}}{\dfrac{a}{(1-a)^{\frac{1}{2}}}}$ $=(1-a)^{\frac{1}{2}}\times\dfrac{(1-a)^{\frac{1}{2}}}{a}$ $=\dfrac{1-a}{a}$ $=\dfrac{1}{a}-1$

NO.	ANSWER	SOLUTION (PART B)
4.	B	$y=\ln\left(\dfrac{3-2x}{5x-1}\right)$ $=\ln(3-2x)-\ln(5x-1)$ $\dfrac{dy}{dx}=\dfrac{-2}{3-2x}-\dfrac{5}{5x-1}$
5.	B	$(3-\log_3 x)\log_{3x}3=1$ $(3-\log_3 x)\left(\dfrac{\log_3 3}{\log_3 3x}\right)=1$ $(3-\log_3 x)\left(\dfrac{\log_3 3}{\log_3 3+\log_3 x}\right)=1$ $(3-\log_3 x)\left(\dfrac{1}{1+\log_3 x}\right)=1$ $3-\log_3 x=1+\log_3 x$ $2\log_3 x=2$ $\log_3 x=1$ $x=3$
6.	C	$f(x)=\dfrac{1}{3}x^3-x^2-8x+2$ $f'(x)=x^2-2x-8$ Critical number: $x^2-2x-8=0$ $(x-4)(x+2)=0$ $x=4 \quad , \quad x=-2$ $f''(x)=2x-2$ $f''(4)=2(4)-2=6>0$ (minimum) $f''(-2)=2(-2)-2=-6<0$ (maximum)

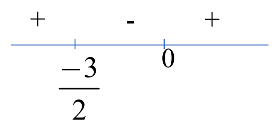
NO.	ANSWER	SOLUTION (PART B)
		When $x=4$: $y=\frac{1}{3}(4)^3-(4)^2-8(4)+2=-\frac{74}{3}$ When $x=-2$: $y=\frac{1}{3}(-2)^3-(-2)^2-8(-2)+2=\frac{34}{3}$ $\therefore$ Minimum point is $\left(4,-\frac{74}{3}\right)$ Maximum point is $\left(-2,\frac{34}{3}\right)$
7.	C	$\lim\limits_{x\to1^+} 3\left(5p^2-4x\right)=\lim\limits_{x\to1^-} 3p^4x^2$ $3\left(5p^2-4\right)=3p^4$ $5p^2-4=p^4$ $p^4-5p^2+4=0$ $\left(p^2-4\right)\left(p^2-1\right)=0$ $p^2=4, p^2=1$ $\therefore p=-1,1,-2,2$
8.	C	$w=\frac{2-\left(1+\left(1-i\right)\right)i}{1+i\left(1-i\right)}$ $=\frac{2-2i+i^2}{1+i-i^2}$ $=\frac{1-2i}{2+i}\times\frac{2-i}{2-i}$ $=\frac{2-5i+2i^2}{4-i^2}$ $=-i$ $\lvert w\rvert=\sqrt{(-1)^2}$ $\qquad\theta=-\frac{\pi}{2}$ $=1$ $w=\cos\left(-\frac{\pi}{2}\right)+i\sin\left(-\frac{\pi}{2}\right)$

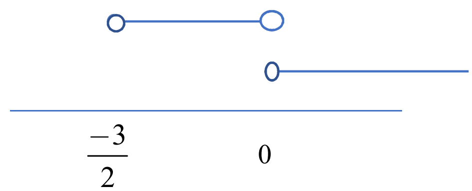
NO.	ANSWER	SOLUTION (PART B)
9.	B	$$4x^4+10x^3y^2=y$$ $$16x^3+10\left[x^3\cdot 2y\frac{dy}{dx}+y^2\left(3x^2\right)\right]=\frac{dy}{dx}$$ $$16x^3+20x^3y\frac{dy}{dx}+30x^2y^2-\frac{dy}{dx}=0$$ $$\frac{dy}{dx}\left[20x^3y-1\right]=-16x^3-30x^2y^2$$ $$\frac{dy}{dx}=\frac{-16x^3-30x^2y^2}{20x^3y-1}$$
10.	D	$$\frac{dx}{dt}=23,\frac{dS}{dt}=?$$ $$\frac{AB}{BD}=\frac{CD}{DE}$$ $$\frac{15}{x+S}=\frac{5.8}{S}$$ $$15S=5.8x+5.8S$$ $$9.2S=5.8x$$ $$S=\frac{29}{46}x$$ $$\therefore\frac{dS}{dx}=\frac{29}{46}$$  $$\frac{dS}{dt}=\frac{dS}{dx}\cdot\frac{dx}{dt}$$ $$=\frac{29}{46}\cdot 23$$ $$=14.5\text{ feet/second}$$
11.	A	$$x=2t^2 \qquad y=2t^5-t^2$$ $$\frac{dx}{dt}=4t \qquad \frac{dy}{dt}=10t^4-2t$$ $$\frac{dy}{dx}=\frac{dy}{dt}\cdot\frac{dt}{dx}$$ $$=\frac{10t^4-2t}{4t}$$ $$=\frac{5}{2}t^3-\frac{1}{2}$$

NO.	ANSWER	SOLUTION (PART B)
		At point $(2,1)$: $2 = 2t^2 \quad , \quad 1 = 2t^5 - t^2$ $t = \pm 1 \qquad \therefore t = 1$ $\frac{dy}{dx} = \frac{5}{2} - \frac{1}{2} = 2$
12.	D	$y = \tan^3(4x)$ $\frac{dy}{dx} = 3\tan^2(4x) \cdot \sec^2(4x) \cdot 4$ $= 12\tan^2(4x)\sec^2(4x)$ When $x = \frac{\pi}{16}$: $\frac{dy}{dx} = 12\tan^2\left(4 \cdot \frac{\pi}{16}\right)\sec^2\left(4 \cdot \frac{\pi}{16}\right)$ $= 24$
13.	B	$\lim_{x \to \alpha^-}(-2x - 2) = \lim_{x \to \alpha^+}(\beta x^2 - 2)$ $-2\alpha - 2 = \beta\alpha^2 - 2$ $-2\alpha = \beta\alpha^2$ -------(1) $\lim_{x \to 5^-} \beta x = \lim_{x \to 5^+}\left(-\frac{2}{x} + \beta\right)$ $5\beta = -\frac{2}{5} + \beta$ $\therefore \beta = -\frac{1}{10}$ $-2\alpha = -\frac{1}{10}\alpha^2$ $-\frac{1}{10}\alpha^2 + 2\alpha = 0$ $\alpha\left(-\frac{1}{10}\alpha + 2\right) = 0$ $\alpha = 0, \quad \alpha = 20$ $\therefore \alpha = 0$

NO.	ANSWER	SOLUTION (PART B)
14.	A	$(h \circ g \circ h)(x) = h(x)$ $= x^3 + 4$ $x^3 + 4 = 12$ $x^3 = 8$ $x = 2$
15.	C	$\dfrac{f(x+h) - f(x)}{h}$ $\dfrac{f(x+h) - f(x)}{h}$ $= \dfrac{\dfrac{4}{1-(x+h)} - \dfrac{4}{1-x}}{h}$ $= \dfrac{4(1-x) - 4(1-(x+h))}{h(1-x)(1-(x+h))}$ $= \dfrac{4}{(1-x)(1-x-h)}$
16.	C	$f \circ g = 2x$ $f\left(e^{2x} - 3\right) = 2x$ let $u = e^{2x} - 3$ $e^{2x} = u + 3$ $2x = \ln(u+3)$ $x = \dfrac{1}{2}\ln(u+3)$ $f(u) = 2\left(\dfrac{1}{2}\ln(u+3)\right)$ $f(u) = \ln(u+3)$ $\therefore f(x) = \ln(x+3)$

NO.	ANSWER	SOLUTION (PART B)
17.	B	$\dfrac{\lvert x+2\rvert}{3\lvert x-4\rvert} \geq 1, x \neq 4$ $\lvert x+2\rvert \geq 3\lvert x-4\rvert$ $(x+2)^2 \geq 3^2(x-4)^2$ $2x^2-19x+35 \leq 0$ $(2x-5)(x-7) \leq 0$ $\therefore \left[\dfrac{5}{2},4\right) \cup (4,7]$
18.	B	$f(x)=\begin{cases} \dfrac{1}{x} & x<\dfrac{1}{3} \\ 6-ax & \dfrac{1}{3} \leq x \end{cases}$ $\lim\limits_{x\to\frac{1}{3}^-}\left(\dfrac{\left(\frac{1}{x}\right)-\left(6-a\left(\frac{1}{3}\right)\right)}{x-\frac{1}{3}}\right)=\lim\limits_{x\to\frac{1}{3}^+}\left(\dfrac{(6-ax)-\left(6-a\left(\frac{1}{3}\right)\right)}{x-\frac{1}{3}}\right)$ $\dfrac{\lim\limits_{x\to\frac{1}{3}^-}\left(\frac{1}{x}-6+\frac{a}{3}\right)}{\lim\limits_{x\to\frac{1}{3}^-}\left(x-\frac{1}{3}\right)}=\dfrac{\lim\limits_{x\to\frac{1}{3}^+}\left[(6-ax)-\left(6-a\left(\frac{1}{3}\right)\right)\right]}{\lim\limits_{x\to\frac{1}{3}^+}\left(x-\frac{1}{3}\right)}$ $\lim\limits_{x\to\frac{1}{3}^-}\left(\frac{1}{x}-6+\frac{a}{3}\right)=\lim\limits_{x\to\frac{1}{3}^+}\left[(6-ax)-\left(6-a\left(\frac{1}{3}\right)\right)\right]$ $\dfrac{1}{\left(\frac{1}{3}\right)}-6+\dfrac{a}{3}=\left(6-a\left(\frac{1}{3}\right)\right)-\left(6-a\left(\frac{1}{3}\right)\right)$ $3-6+\dfrac{a}{3}=0$ $\dfrac{a}{3}=3$ $a=9$
19.	C	$\left\lvert\dfrac{x+3}{x}\right\rvert>1$ $\dfrac{x+3}{x}>1$ or $\dfrac{x+3}{x}<-1$

<table>
<tr><th>NO.</th><th>ANSWER</th><th>SOLUTION (PART B)</th></tr>
<tr><td></td><td></td><td>
$\frac{x+3}{x}-1>0$
$\frac{x+3-x}{x}>0$
$\frac{3}{x}>0$
$\therefore x>0$
or
$\frac{x+3}{x}+1<0$
$\frac{x+3+x}{x}<0$
$\frac{2x+3}{x}<0$


$\therefore \frac{-3}{2}<x<0$


$\therefore \left(\frac{-3}{2},0\right)\cup(0,\infty)$
</td></tr>
<tr><td>20.</td><td>D</td><td>
$x^2+y^2=\frac{y}{x}$
$x^3+xy^2=y$
$3x^2+y^2+2xy\frac{dy}{dx}=\frac{dy}{dx}$
$3x^2+y^2=(1-2xy)\frac{dy}{dx}$
$\frac{dy}{dx}=\frac{3x^2+y^2}{(1-2xy)}$
</td></tr>
<tr><td>21.</td><td>B</td><td>
$f^2\left(g(-2)\right)=ff(1)=f\left(f(1)\right)=f(0)=1$
$gf\left(g(-1)\right)=gf(0)=g(1)=-2$
</td></tr>
</table>

NO.	ANSWER	SOLUTION (PART B)
22.	C	$g\left(f\left(x\right)\right)=e^{x}$ $g\left(e^{2x}\right)=e^{x}$ Let $m=e^{2x}$, $2x=\ln m$ $x=\frac{1}{2}\ln m$ $=\ln\sqrt{m}$ $g\left(m\right)=e^{\ln\sqrt{m}}$ $=\sqrt{m}$ $\therefore g\left(x\right)=\sqrt{x}$
23.	C	$D_{f^{-1}}=\left(0,\frac{7}{2}\right),R_{f^{-1}}=\left[\frac{-9}{2},0\right)\cup\left(0,1\right]$
24.	D	$g:x\to\left(x-1\right)^{2}+2,x\in\left[0,\infty\right)$ $g\left(g^{-1}\left(x\right)\right)=x$ $\left(g^{-1}\left(x\right)-1\right)^{2}+2=x$ $g^{-1}\left(x\right)-1=\sqrt{x-2}$ since $x\geq 0$ $g^{-1}\left(x\right)=\sqrt{x-2}+1$ $D_{g^{-1}}=\left[2,\infty\right)$

NO.	ANSWER	SOLUTION (PART B)
25.	D	The volume of the cube increases as it is heated. Side increases at the rate of 0.005cms^{-1}. $V = x^3$ $\frac{dV}{dx} = 3x^2$ $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$ $= 3x^2 \times 0.005$ $= 0.015x^2$ $t = 20,\ x = 9.9 + 20(0.005) = 10$ When Hence, $\frac{dV}{dt} = 0.015(10)^2$ $= 1.5\text{cm}^3\text{s}^{-1}$